Jiayi Li

jiayi.li.math@gmail.com · jl2ml.github.io · Google Scholar · ORCID 0000-0001-5132-7896

Profile

I study the geometry and statistics of neural networks: the functions an architecture can represent, the solutions that training selects, and what can be identified from data. The tools are algebraic geometry, singular learning theory, and statistics. The aim is a mathematical foundation for interpretability and AI alignment: statements about learned systems that hold as theorems under stated hypotheses.

Employment & Affiliations

Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)
Dresden, Germany
Section of Mathematics and Artificial Intelligence 2025–present
Postdoctoral researcher

ILIAD
Fellow, applied mathematics for AI alignment Sep–Dec 2026

Education

University of California, Los Angeles 2025
Ph.D. in Statistics, research focus in machine learning theory
Thesis: *Structure, Symmetry, and Singularity in Learning Systems*

The University of Hong Kong & Stony Brook University 2018
B.S. in Mathematics, honors thesis in algebraic geometry
Completed the first two years of the math PhD course sequence

Honors, Fellowships & Grants

ILIAD Fellowship, applied mathematics for AI alignment	2026
Conference grant, German Research Foundation (EUR 4,000)	2026
Dimitris N. Chorafas Foundation Prize (USD 10,000)	2024
Dissertation Year Fellowship, UCLA (USD 36,000)	2024
Distinguished Teaching Award Nominee, Statistics/MSOL, UCLA	2024
Summer Mentored Research Fellowship, UCLA (USD 6,000)	2022
ACM Scholarship, Association for Computing Machinery (USD 2,000)	2020
Cathay Bank Scholarship (USD 6,000)	2020
Overseas Research Fellowship, HKU (HKD 12,000)	2016
Undergraduate Research Fellowship, HKU (HKD 15,000)	2015
Travel grants: 20+ awards, over USD 25,000 cumulative	2017–2025

Preprints

9. **Jiayi Li**. "*The Statistical Cost of Redundant Width in Bayesian Quadratic Neural Networks*", under review, 2026; manuscript available upon request
8. Ben Cullen, Sergio Estan-Ruiz, Riya Danait, **Jiayi Li**. "*A Basin-Selection Perspective on Grokking via Singular Learning Theory*", arXiv:2603.01192, 2026
7. Cecilie Olesen Recke, Sarah Lumpp, Nataliia Kushnerchuk, Janike Oldekop, **Jiayi Li**, Jane Ivy Coons, Elina Robeva. "*Identifiability in Graphical Discrete Lyapunov Models*", arXiv:2601.21818, 2026; minor revision at *Electronic Journal of Statistics*

Selected Publications

6. Yulia Alexandr, Miles Bakenhus, Maize Curiel, Sameer K. Deshpande, Elizabeth Gross, Yuqi Gu, Max Hill, Joseph Johnson, Bryson Kagy, Vishesh Karwa, **Jiayi Li**, Hanbaek Lyu, Sonja Petrović, Jose Israel Rodriguez. "*New Directions in Algebraic Statistics: Three Challenges from 2023*", Algebraic Statistics, 15, 357–382, 2024
5. Kaie Kubjas, **Jiayi Li**, Maximilian Wiesmann. "*Geometry of Polynomial Neural Networks*", Algebraic Statistics, 15, 295–328, 2024

4. Shuang Liang, Renata Turkeš, **Jiayi Li**, Nina Otter, Guido Montúfar. "*Pull-back Geometry of Persistent Homology Encodings*", Transactions on Machine Learning Research (TMLR), 2024
3. **Jiayi Li**, Yuantong Li, Xiaowu Dai. "*Discussion of 'Estimating Means of Bounded Random Variables by Betting' by Waudby-Smith and Ramdas*", Journal of the Royal Statistical Society: Series B (JRSSB), 86(1), 41–43, 2024
2. Dejun Guo, Xu Jin, Dan Shao, **Jiayi Li**, Yang Shen, Huan Tan. "*Image-Based Regulation of Mobile Robots without Pose Measurements*", IEEE Control Systems Letters (L-CSS), vol. 6, pp. 2156–2161, 2022
1. Ziqi Huang, Yang Shen, **Jiayi Li**, Marcel Fey, Christian Brecher. "*A Survey on AI-Driven Digital Twins in Industry 4.0: Smart Manufacturing and Advanced Robotics*", Sensors, 21(19), 6340, 2021

Professional Service

Editorial and committees

Member, Assessment & Search Committee, ACM Publications Board 2026–present
Search committee liaison for the magazines group

Editor-in-Chief, XRDS: Crossroads (ACM magazine for students) 2022–2025
Lead Editor 2021; Feature Editor 2020

Program committees

Area Chair, NeurReps 2026: Symmetry and Geometry in Neural Representations, NeurIPS 2026 Workshop, Sydney, Australia 2026

Conference organization

Co-organizer, Singular Learning Theory Days (two-day international workshop)
MPI-CBG, Dresden, Germany Oct 26–27, 2026

Co-organizer, AMS Special Sessions at the Joint Mathematics Meetings 2025–2027
Algebraic Methods in Machine Learning and Optimization
JMM 2025, Seattle; JMM 2026, Washington D.C.
Algebraic, Topological and Geometric Methods in Machine Learning
JMM 2027, Chicago

Minisymposium co-organizer, SIAM Conference on Mathematics of Data Science (MDS26)
Salt Lake City, UT, USA Nov 16–20, 2026

Reviewing

NeurIPS, ICML, ICLR, Algebraic Statistics, SIAM Journal on Applied Algebra and Geometry, Effective Methods in Algebraic Geometry (MEGA)

Mentoring

Co-mentor with Martin Trapp, London Geometry and Machine Learning Summer School (LOGML) 2026 Jul 13–17, 2026
Project: *Posterior Geometry and Variational Inference for Bayesian Complexity in Neural Networks*

Mentor, LOGML 2025 Jul 7–11, 2025
Project: *Symmetry, Degeneracy and Effective Dimensions of Neural Networks*
The project developed into the preprint arXiv:2603.01192

Academic mentor, IPAM Research in Industrial Projects for Students (RIPS)
Project: *Geometry of Representation in Large Language Models* Jun–Aug 2025
Industrial partner: Microsoft Research

Research Visits

Institute for Computational and Experimental Research in Mathematics (ICERM)
Providence, RI, USA Jan 20–Apr 23, 2027
In-residence participant, *Semester Program: Metric Algebraic Geometry — Going Global*

Institute for Pure & Applied Mathematics (IPAM)
Los Angeles, CA, USA Aug 31–Sep 4, 2026
Workshop participant (poster and lightning talk), *Foundations of Interpretability*

Max Planck Institute for Mathematics in the Sciences (MPI MiS)
Leipzig, Germany Summers 2024 and 2025
Visiting Graduate Researcher, Group: Mathematical Machine Learning

Institute for Pure & Applied Mathematics (IPAM)
Los Angeles, CA, USA Sep–Dec 2024
Graduate Fellow, *Long Program: Mathematics of Intelligences*

Simons Institute for the Theory of Computing
Berkeley, CA, USA Fall 2024
Short-Term Visitor, *Modern Paradigms in Generalization*

Institute for Mathematical & Statistical Innovation (IMSI)
Chicago, IL, USA Sep–Dec 2023
Graduate Fellow, *Algebraic Statistics and Our Changing World*

Invited Talks (24)

2026

24. Machine Learning in Life Science (MLLS) Seminar
Oct 2, 2026, Pioneer Centre for AI, Copenhagen, Denmark
"Geometry and Statistical Complexity of Neural Networks"

23. Mathematics Colloquium, Nanjing University
Aug 24, 2026, Nanjing, China
"Structure, Symmetry and Singularity in Modern Neural Networks"

22. Academy of Mathematics and Systems Science, Chinese Academy of Sciences
Aug 11, 2026, Beijing, China
"Structure, Symmetry and Singularity in Neural Networks"

21. Machine Learning Seminar, École Normale Supérieure
May 21, 2026, Paris, France
"Structure, Symmetry and Singularity in Modern Neural Networks"

20. Conference on Geometry, Topology and Machine Learning, Sorbonne Université
May 18–22, 2026, Paris, France
"Basin Selection in Bayesian Learning"

19. Applied CATS Seminar, KTH Royal Institute of Technology
Mar 3, 2026, Stockholm, Sweden
"Grokking as a Phase Transition Between Competing Basins, a Singular Learning Theory Perspective"

2025

18. Mathematical Foundations of AI, Lie–Størmer Colloquium
Dec 3–5, 2025, Simula Research Laboratory, Oslo, Norway
"Structure, Symmetry and Singularity in Learning Systems"

17. Math & Machine Learning Seminar, Boston College
Oct 15, 2025, online
"Theoretical Insights for Optimizing Rational Activations"

16. Applied Algebra and Geometry UK Network, Edinburgh meeting
Aug 21, 2025, Edinburgh, UK
"Geometry of Neural Networks with Algebraic Activations"

15. Algebraic Methods in Optimization, ICCOPT
Jul 21–24, 2025, Los Angeles, CA, USA
"Optimizing Rational Neural Networks with Trainable Parameters"

14. SIAM Conference on Applied Algebraic Geometry
Jul 7–11, 2025, Madison, WI, USA
"Geometry of Neural Networks with Algebraic Activations"

13. Mathematics, AI and Data Science for Material Innovations (MADSMIN), Lancaster University
Jun 9–13, 2025, Lancaster, UK
"Geometry of Neural Networks with Algebraic Activations"

12. Geometry and Machine Learning (GML2025), Sorbonne Université
May 26–30, 2025, Paris, France
"Geometry of Neural Networks with Algebraic Activations"

11. Workshop on the Statistical Theory of Neural Networks
May 5–8, 2025, Egmond aan Zee, Netherlands
"Geometry of Neural Networks with Algebraic Activations"

10. Numerical (Nonlinear) Algebra in the Real World
Feb 6, 2025, Dresden, Germany
"Taming Non-convexity in Shallow Neural Networks with Algebraic Activations"

9. Joint Mathematics Meetings (JMM) 2025
AMS–ASA–SIAM Special Session on Mathematics of Deep Learning: A High-Dimensional Probability Perspective
Jan 11, 2025, Seattle, WA, USA
"Algebro-geometric Approaches to Optimization and Generalization in Mathematical Machine Learning"

2024

8. Algebraic Geometry and Machine Learning Workshop, SIAM Conference on Mathematics of Data Science
Oct 25, 2024, Atlanta, GA, USA
"Geometry of Polynomial Neural Networks"

7. Numerical Analysis Seminar, Institute of Mathematical Research and Dept. of Mathematics, The University of Hong Kong
Apr 30, 2024, Hong Kong
"Geometry of Polynomial Neural Networks"

6. Special Session: Applications of Algebra and Geometry, American Mathematical Society 2024 Spring Central Sectional Meeting
Apr 20, 2024, Milwaukee, WI, USA
"An Algebraic Approach to Supply Network Formation and Fragility"

5. Dept. of Mathematics and Statistics, University of Massachusetts Amherst
Apr 12, 2024, online
"Geometry of Polynomial Neural Networks"

4. Level Set Seminar, Dept. of Mathematics, UCLA
Apr 8, 2024, Los Angeles, CA, USA
"Geometry of Polynomial Neural Networks"

2023

3. Applied Algebra Seminar, Dept. of Mathematics, University of Wisconsin–Madison
Dec 2023, Madison, WI, USA
"Geometry of Polynomial Neural Networks"

2. Deep Learning Theory Seminar, UCLA
Sep 2023, Los Angeles, CA, USA
"Neural Collapse Beyond Unconstrained Feature Models"

1. SCISS, UCLA
Jul 2023, Los Angeles, CA, USA
"Mathematical Machine Learning: Theory & Beyond"

Contributed Talks (6)

6. Workshop on Real Applied Algebra
Mar 23–25, 2026, University of Copenhagen, Denmark
"LLC in Transformers: An Algebro-Geometric Lens on Generalization in Neural Networks"

5. Higher-order interactions at the crossroads of geometry and topology
Dec 10–12, 2025, Paris, France
"Geometry of Higher-Order Structures in Machine Learning"

4. Matroids, Rigidity, and Algebraic Statistics, ICERM Workshop
Mar 17, 2025, Providence, RI, USA
"Geometric Framework for Optimizing Algebraic Neural Networks"

3. 2024 UC Davis Peter Hall Conference on Statistics in the Age of AI
Nov 9, 2024, Davis, CA, USA
"Optimization in Polynomial Neural Networks: Insights from Algebraic Geometry"

2. Women in Algebraic Statistics Workshop, St John's College, University of Oxford
Jul 12, 2024, Oxford, UK
"Algebraic Structures in Terminal Phase of Neural Network Training"

1. Institute for Mathematical and Statistical Innovation (IMSI)
Oct 10, 2023, Chicago, IL, USA
"Simplex ETF – Algebraic structure arising in the terminal phase of neural network training"

Teaching

Invited minicourse: *Singular Learning Theory*, within the program *An Introduction to the Mathematical Foundations of AI Safety*
SISSA, Trieste, Italy — Nov 23–27, 2026

Summary: Instructor, Teaching Associate, and Teaching Assistant for 25+ course offerings across statistics, engineering, and machine learning, reaching 1000+ students at UCLA and the Oxford Machine Learning Summer School.

Instructor of record: Prep Camp, Master of Applied Statistics & Data Science (MASDS), UCLA, 2021–2023

Teaching Associate / Teaching Assistant, UCLA, 2020–2023:
Statistics and Data Science (STATS 13, ENGR 116)
Engineering Systems (ENGR 200, 202, 203, 205)

Teaching Assistant, Oxford Machine Learning Summer School

Outreach

Founding organizer, Distinguished Women in Statistics and Data Science Workshop Series, UCLA: 1st (Women in Academia, May 23, 2023) and 2nd Symposium (Women in Industry, Oct 17, 2023)
UCLA Society of Women in Statistics (Committee, 2022–2024)
UCLA Statistics Club (Mentor, 2022–2024)
Invited speaker: Rotary Club, Westwood Village (2023); UCLA Bruin Professionals (2024); Dublin High School, Dublin, CA (2024)

Memberships

American Mathematical Society (AMS), Association for Women in Mathematics (AWM), American Statistical Association (ASA), Institute of Mathematical Statistics (IMS), Society for Industrial and Applied Mathematics (SIAM), Association for Computing Machinery (ACM)